AIORI-3 HACKATHON LEVEL 1 EVALUATION PROPOSAL

Problem Statement: PS-019: Personal-Data Discovery and Privacy Evidence Inventory
Category: Cybersecurity | **Framework:** MeitY DPDPA 2023 Aligned

Executive Overview: Under modern privacy regimes like India's Digital Personal Data Protection Act (DPDPA 2023), organizations must catalog all stored Personally Identifiable Information (PII) and bind it to declared purposes and statutory retention schedules. Unindexed transactional databases and flat spreadsheets frequently accumulate 'shadow personal data' without documentation. This project presents a technical discovery scanner and evidence inventory engine that inspects synthetic CSVs and structured SQL databases, classifies personal-data across five classes with **95.4% Macro F1-score**, maps governance metadata, and unmasks compliance gaps without overstepping into automated legal interpretations.

1. Problem Description

Modern enterprise systems continuously ingest user information across operational pipelines. Over time, personal data accumulates in unindexed database tables and ad-hoc CSV exports without audit trails. Organizations face substantial statutory liabilities if they fail to identify stored PII or hold data beyond legitimate necessity (storage limitation). However, technical scanners must strictly distinguish objective technical evidence from legal compliance decisions: automated systems must NOT make legal lawful-basis determinations or infer universal cross-border flows. The challenge is to engineer an accurate, configurable technical discovery scanner that maps data across 5 distinct classes, links fields to corporate retention policies, and surfaces evidence gaps for human DPO sign-off.

2. Solution Proposed

We have developed a comprehensive discovery and privacy evidence inventory suite supporting both multi-table SQL databases (SQLite) and flat CSV datasets. The solution incorporates:

• **Five Predefined Data Classes:** (1) Direct Personal Identifiers, (2) Contact & Location Data, (3) Government National IDs (Aadhaar, PAN, Passport), (4) Financial & Banking Data (Cards, Bank Acc, IFSC, UPI), and (5) Digital Identity & Telemetry (IP, MAC, Hashes, Session UUIDs).
• **Purpose & Retention Linker:** Connects discovered columns with declared corporate privacy policies (governance_registry.json), mapping business purposes, retention periods, and statutory citations.
• **Evidence Gap & Shadow Data Flags:** Flags fields with missing purposes (GAP_MISSING_PURPOSE) or missing/indefinite retention (GAP_MISSING_RETENTION).
• **Legal Safeguard Boundaries:** Explicitly classifies output as 'Technical Evidence Inventory' with mandatory disclaimers.

3. Optimization Proposed by the Team

To avoid the fragility of pure column-name heuristics and the computational overhead of full-table scanning, our team engineered a **Two-Tier Hybrid Classification Engine with Cross-Validation Boosting**:

• **Header Semantic Normalization (Tier 1):** Evaluates regular expressions against normalized column headers ($W_h = 0.50$).
• **Deep Sample-Value Regex Inspection (Tier 2):** Samples up to 100 non-null values and runs specialized patterns ($W_v = 0.50$), including Luhn heuristics, Verhoeff Aadhaar patterns, Indian IFSC formats, and RFC-5322 email syntax.
• **Cross-Validation Bonus (+0.10):** If both header syntax and actual sample values match, confidence is boosted.
• **Negative Specificity Safeguard (93.94%):** Prevents false alarms on generic business numbers (SKU, price, stock quantity).

4. Solution Architecture and Design

The architecture decouples data ingestion, pattern classification, governance policy mapping, and benchmark evaluation:

Module / Layer	Component Role	Implementation Files
Data Ingestion	Introspects SQL tables and flat CSVs, extracts 100 sample rows	core/scanner.py, data/raw/, enterprise.db
Rules Engine	Two-tier header and sample pattern evaluation across 5 classes	core/rules_engine.py, config/rules_config.json
Metadata Linker	Maps declared purpose, retention schedules, and flags shadow PII	core/metadata_linker.py, config/governance_registry.json
Evaluation Suite	Calculates TP, FP, FN, Precision, Recall, F1 & Confusion Matrix	core/evaluator.py, data/benchmark/ground_truth_labels.json
Presentation Layer	Interactive Flask Web Dashboard and Rich Terminal CLI	web/app.py, web/templates/, cli.py

Evaluation Benchmark Performance (66 Columns Evaluated):

Data Class	TP	FP	FN	Support	Precision	Recall	F1-Score
Direct Personal Identifiers	7	0	0	7	100.0%	100.0%	100.0%
Contact & Location Data	10	1	0	10	90.91%	100.0%	95.24%
Government & National IDs	2	0	0	2	100.0%	100.0%	100.0%
Financial & Banking Data	5	0	1	6	100.0%	83.33%	90.91%
Digital Identity & Telemetry	8	2	0	8	80.00%	100.0%	88.89%
Macro Average / Summary	32	3	1	33	94.18%	96.67%	95.41%
Non-PII Negative Specificity	--	--	--	33	--	--	93.94%

5. Implementation Details & Prototype Demonstration (Pseudocode)

Core discovery, value validation, and governance mapping pseudocode for mentor evaluation:

MeitY DPDPA 2023 Alignment: This technical suite implements technical evidence discovery aligned with the MeitY Data Protection Framework (https://www.meity.gov.in/data-protection-framework). It creates objective evidence records without making automated legal determinations, empowering Data Protection Officers to conduct defensible compliance audits.